

Precisely Priced

The TUR 2:1 Floor and the Demon's Handling Fee

Knowing Costs Twice: Accuracy, Then Erasure

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L.O.R.E. Framework (Law of Repulsive Emanation)

Abstract

The thermodynamic uncertainty relation (Barato-Seifert 2015; Gingrich et al. 2016; time-dependent form Dechant-Sasa 2018) sets a 2:1 price floor on the ledger: for a cyclic $\Delta F = 0$ process with $\langle W \rangle > 0$, $\text{Var}(W) \geq 2 \langle W \rangle$. We measure the slack $s = 1 - q$, $q = 2 \langle W \rangle / \text{Var}$, on two instruments. The dragged Gaussian trap (work exactly Gaussian, Mazonka-Jarzynski 1999) saturates the bound: $q = 0.982-0.993$ with skew = kurt = 0 to measurement error - the slack is a discretization residue, vanishing toward the continuum. The stiffness round trip of Ch.74/75 (work non-Gaussian, skew +1.25, kurt +6.6) pays slack 0.105; the demon's coin bends it further - dead sign bit slack 0.159, engaged far/near bit slack 0.54 $\rightarrow$ 0.89 as the leverage grows and q collapses to 0.106. Harvest rows ($\langle W \rangle < 0$) leave the $X > 0$ province of the TUR, where Ch.75's erasure bill governs ($W_{\text{net}} = +0.656-0.670$). Knowing costs twice: once in precision (this chapter), once in erasure (Ch.75).

1. The 2:1 Floor

For any time-integrated current or accumulated observable X with $\langle X \rangle > 0$ and total entropy production Σ ,

$$\text{Var}(X) / \langle X \rangle^2 \geq 2 / \langle \Sigma \rangle \quad (k_B T = 1)$$

On a cyclic round trip with $\Delta F = 0$ the entropy production is $\langle \Sigma \rangle = \langle W \rangle$ exactly (the initial and final potentials coincide), so the bound reads:

$$\text{Var}(W) \geq 2 \langle W \rangle \quad ; \quad q = 2 \langle W \rangle / \text{Var}(W) \leq 1 \quad ; \quad \text{slack } s = 1 - q \geq 0$$

The slack is the price of any deviation of the work distribution from the parabola. Ch.72 measured the same constant from the other side ($\text{Var}/(2\langle W \rangle) \sim 1$): the TUR is the fluctuation-dissipation tie raised to a theorem.

2. The Parabola Saturates

Instrument A: dragged fixed-stiffness trap $V = k(x - \lambda)^2/2$, $k = 2$, $\lambda \rightarrow 0 \rightarrow 2 \rightarrow 0$, $D = \beta = 1$, $\Delta F = 0$. The work is a linear functional of the Gaussian bath, hence exactly Gaussian (Mazonka-Jarzynski 1999, Onsager-Machlup 1953); the Gaussian Jarzynski equality $\langle e^{-W} \rangle = 1$ forces $\text{Var}(W) = 2 \langle W \rangle$ exactly in the continuum. Measured (150,000 Heun SRK2 runs per row):

tau	$\langle W \rangle$	Var	q	slack	skew	kurt
1.0	+8.9815	18.28835	0.9822	0.0178	+0.0030	-0.025
2.0	+5.9852	12.04974	0.9934	0.0066	-0.0006	-0.009
4.0	+3.4627	7.02915	0.9852	0.0148	-0.0019	+0.016

Skew and excess kurtosis are zero within their 0.006 and 0.013 standard errors on every row: the work is Gaussian, and Gaussian saturates. The residual slack (1-2 CE) is finite-step $O(dt)$ discretization, vanishing as $DT \rightarrow 0$. Precision squared $\langle W \rangle^2 / \text{Var}$ equals $\langle \Sigma \rangle / 2$ to the same accuracy: the parabola knows exactly what it pays.

3. The Fee Is the Bending

Instrument B (the demon of Ch.74/75): stiffness round trip $V = \lambda x^2/2$, $\lambda \rightarrow 2 \rightarrow 1$, the fair far/near bit at the median of $|x_{\text{mid}}|$ (median 0.5647). The stiffness work rides on x^2 , so its distribution is already non-Gaussian - control skew +1.25, kurt +6.6 - and the bound is strict. The coin bends it further:

row	$\langle W \rangle$	Var	q	slack	skew	kurt	ln J
control 0.5	+0.11263	0.25167	0.8951	0.1049	+1.250	+6.642	+0.001
dead sign 0.4/3	+0.11095	0.26377	0.8413	0.1588	+1.708	+8.018	-0.002
engaged 0.4/1.5	+0.06114	0.26492	0.4616	0.5384	+1.235	+6.779	+0.058
engaged 0.35/2	+0.04476	0.25504	0.3510	0.6490	+1.168	+5.762	+0.070
engaged 0.25/4	+0.01279	0.24172	0.1058	0.8942	+1.217	+6.676	+0.097

The between-group variance of the mixed return speeds is pure waste: q collapses from 0.895 to 0.106 as the demon works harder, and even the dead coin (no information) pays a fee (0.841). The ledger then does exactly what the demon wants (ln J rises to 0.097) while the precision budget drains: using information is expensive in accuracy, before any erasure bill is paid.

4. The Harvest Province

At the strongest leverage the mean work goes negative (the demon harvests, Ch.74) and the TUR's $\langle X \rangle > 0$ province is exited: q is formally undefined, and the Ch.75 erasure bill governs instead.

$$\text{HARVEST } 0.10/8: \quad \langle W \rangle = -0.02303, \quad \ln J = +0.11930, \quad W_{\text{net}} = +0.67012$$

$$\text{HARVEST } 0.05/16: \quad \langle W \rangle = -0.03729, \quad \ln J = +0.12571, \quad W_{\text{net}} = +0.65586$$

The harvest is capped by the coin (Ch.75); the precision it costs is counted by the slack it leaves behind. Two prices, one budget: accuracy first, then erasure.

5. The 0/0: the Removable Value of Knowing Per Dissipation

Precision squared $P^2 = \langle W \rangle^2 / \text{Var}$ and dissipation per unit time $\Sigma/2 = \langle W \rangle / 2$ vanish together in the reversible limit - a 0/0 whose ratio is the TUR ratio:

$$P^2 / (\Sigma/2) \rightarrow q \rightarrow 1 \quad \text{on the parabola} \quad (\text{removable value } 1)$$

The removable value of "knowing per dissipation" is 1, reached only by Gaussian (linear, silent) reading; every non-linearity - a stiffness ramp, a coin, a demon - bends the parabola and pays a positive fee $1 - q$, measured from 0.10 to 0.89 here. The TUR is therefore the bookkeeper of the 0/0: it prices the removable value of precision and bills every deviation from the tangent.

Main Results

THEOREM

The 2:1 Floor: For a cyclic round trip ($\Delta F = 0$) with $\langle W \rangle > 0$, $\text{Var}(W) \geq 2 \langle W \rangle$. Gaussian work (a linear dragged trap) saturates the bound in the continuum - measured $q = 0.982$ - 0.993 with skew = kurt = 0, the slack a pure discretization residue.

COROLLARY

The Slack Is the Deviation Fee: the non-Gaussian stiffness round trip pays slack 0.105; the demon's engaged bit drives it to 0.54-0.89 (q to 0.106) - using information is expensive in accuracy.

COROLLARY

The Dead Coin Also Fees: a bit uncorrelated with the work (sign $\rightarrow$ speed) still lowers q (0.895 $\rightarrow$ 0.841): silence is not free.

COROLLARY

The Harvest Province Is the Erasure Province: where $\langle W \rangle < 0$ the TUR releases the books to Ch.75's bill ($W_{\text{net}} = +0.656$ - 0.670): knowing costs twice, precision then erasure.

Conclusion

Chapter 75 priced the coin's erasure; this chapter prices the act of knowing itself. The TUR floor $\text{Var} \geq 2 \langle W \rangle$ is met exactly by Gaussian work - the removable value 1 of knowing per dissipation - and every departure bills itself: a stiffness ramp pays 0.10, a dead coin 0.16, an engaged coin up to 0.89. The demon's total account is two lines, one in accuracy and one in erasure, and both are debited before any harvest is credited. Books close with the coin, and the coin is metered by variance.

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